

CHAPTER 6

INTERTIDAL ECOLOGY

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

Although the intertidal zone is minuscule when compared with the rest of the marine environment, it possesses a rich biota. Its importance in educating the public about the marine environment cannot be overemphasized, for it is the intertidal shore that most people envision when they think of the ocean. News reports of helpless waterfowl coated with oil or medical trash washing up on the beach present powerful images that can be used to educate the public about the importance of protecting this region of our environment (see also Chapter 11).

This long chapter is divided into seven sections. The first, ENVIRONMENTAL CONDITIONS, describes the general habitats to be discussed, while the second, ADAPTATIONS OF INTERTIDAL ORGANISMS, reviews the adaptations of the organisms living there. The next three sections cover the habitats of this environment: ROCKY SHORES; SANDY SHORES; MUDDY SHORES. The last two sections explore animals not mentioned in previous chapters: INTERTIDAL FISHES; BIRDS.

I. Environmental Conditions

Wasn't it the Scarecrow in *The Wizard of Oz* who yearned to answer the profound question about the relative juxtaposition of the ocean and the shore?—*Why is the ocean close to the shore?* This section, which describes the habitats of the intertidal shore, is divided into four parts:

- Tides
- Temperature
- Wave Action
- Salinity

23.5°	ice	tidal range
apogee	mixed tide	tidal zone
centrifugal force	neap tide	tide
critical tide level (CTL)	osmosis	tide pool
diurnal tide	perigee	wave action
duration of exposure	semidiurnal tide	
gravitational attraction	spring tide	

1. The general concept of tides is easily understood by students, but the variations of diurnal, semi-diurnal, and mixed tides sometimes confuse a few of them, especially students who have never experienced a long stay at the shore. Students may ask, "Why isn't there a single tide throughout all the oceans?" Offering the statement that basin shape influences tidal action generally does not help these students to understand the concept.

One approach to overcome this difficulty is to fit a constructed coastline of bays and islands into a large shallow basin (at least 0.3 × 3 m) filled with water. A child's wading pool works well, but no matter what sort of basin you use, the arrangement of the fake coastline (typically constructed of rocks, bricks, and cinder blocks) takes some trial-and-error work to perfect. Three or four students act as observers or tide watchers. Then subject this world to a periodic tide—really, a slow, periodic wave train produced by raising and lowering a small bucket at one end. What happens, of course, is that some of the waves interact with each other, reinforcing here and canceling there, so that the tide observers report different types of tides.

Duplicating the Bay of Fundy is not easily done. In addition, this educational operation is time consuming; however, it does help illustrate the concept of tides. Usually, I have less time available than needed for the demonstration, so I discuss a somewhat analogous situation instead: the flow of cars in a traffic jam. This analogy is something that almost everyone can appreciate.

2. The critical tide levels represent closely spaced regions of the shore where movement of only a few centimeters above or below those points can significantly influence the average daily length of time of exposure of the shore to the air and, therefore, to desiccation and extremes in temperature. Taking the opposite point of view, one might rephrase this statement to emphasize the length of submergence and, therefore, the amount of time available for organisms to feed. Exposure, temperature, wave action, salinity, and substratum type are very important in determining the organisms present in an intertidal region. The text covers this topic quite well, but you may wish to have your students examine the original literature (Stephenson & Stephenson, 1949) as cited in the text.

II. Adaptations of Intertidal Organisms

Most of the adaptations seen in intertidal organisms involve the ability to deal with the daily rise and fall of the tides and the problems that the tides bring (exposure, wave action, etc.). This section is divided into seven parts:

- Resistance to Water Loss
- Maintenance of Heat Balance
- Mechanical Stress
- Respiration
- Feeding
- Salinity Stress
- Reproduction

antifreeze
byssal threads
cutaneous respiration
desiccation
gills
heat balance

home scar
mantle cavity
neap tides
operculum
osmoconformers
osmoregulators

osmosis
radiation
refuge
seaweed canopy
spring tides
streamline

1. Desiccation is a serious problem that must be overcome by intertidal organisms. Animals migrate with the tides, find or excavate burrows, hide under moist algae, or rely on a shell that is impermeable to water. Some plants can endure severe water loss and still be revived.
2. Overheating in the summer and freezing in the winter are also problems for intertidal organisms. All adaptations to deal with heat problems involve two simple concepts: reducing heat gain and increasing heat loss.
3. Wave action can be very harsh on organisms living in the intertidal zone. Some organisms overcome this physical stress by gluing themselves to a firm substratum—that is, by being sessile (for example, barnacles and oysters). Others possess strong shells that can deflect much of the wave action (for example, limpets and chitons).
4. At low tide on hot summer days, the water remaining in the tide may evaporate somewhat, thereby elevating the local salinity. Organisms respond to these dangers by being either osmoconformers or osmoregulators.

III. Rocky Shores

This long section covers a region well loved by the general public and marine biologists alike. It consists of 10 subsections:

- Atlantic and Pacific Rocky Shores
- Zonation
- Causes of Zonation
- Patchiness
- Succession
- Horizontal Distribution Patterns
- Tide Pools

- Tropical Intertidal Shores
- Cobble Beaches
- Age Structure

acrorhagi	fugitive species	patch
age class	grazing	pioneer species
age structure	habitat-dependent selection	predation
age-specific survival	habitat-dependent survival	propagules
allelochemic	habitat facilitation	<i>r</i> - vs. <i>K</i> -selection
<i>Balanus</i>	habitat modification	recruitment
barnacle zone	halogenated metabolites	refuge
barnacles	herbivory	resource monopolization
bet hedging	home scar	space holder
caging experiments	infralittoral fringe	spray zone
chemical defenses	insolation	Stephenson Universal Scheme
<i>Chthamalus</i>	Irish moss zone	sublittoral zone
clonal	kelp zone	substrate stabilization
clone	keystone predator	succession
community structure	keystone species	supralittoral fringe
competition	larval recruitment	temperature
critical tide levels (CTL)	larval settlement	territoriality
crusting growth form	littoral zone	tide pool
desiccation	local distribution	<i>Torrey Canyon</i>
diffuse predation	logs	UV light
El Niño	midlittoral zone	water seep
encrusting	mussel zone	weak predation
frondose growth form	opportunistic species	zonation

1. The harlequin pattern of color and texture displayed by zonation of organisms on rocky shores is indeed striking and very reminiscent of mountain zonation, albeit on a much finer scale. Much research has employed the efforts of many scientists to elucidate these patterns.
2. The Stephenson Universal Scheme was an attempt to standardize the cacophony of terms invented by workers to describe patterns of zonation distinct to specific locales. The Stephensons' paper is referenced at the end of the chapter.
3. Physical factors, such as critical tide levels and water seeps, and biological factors, such as competition, predation, grazing, and larval recruitment (settlement), interact to produce the zonation patterns of rocky coasts. A significant research effort has been directed at understanding zonation. The textbook provides several excellent examples of classical studies, including the following:

Denny, M.W., et al. 2003. Extreme water velocities: topographical amplification of wave-induced flow in the surf zone of rocky shores. *Limnol. Oceanogr.* 48: 1–7.

Doty, M.S. 1946. Critical tide factors that are correlated with the vertical distribution of marine algae and other organisms along the Pacific coast. *Ecology* 27: 315–328.

Newell, R.C. 1979. *Biology of intertidal animals*. Faversham, Kent, U.K.: Marine Ecological Surveys Ltd.

Somero, G.N. 2002. Thermal physiology and vertical zonation of intertidal animals: optima, limits, and costs of living. *Integr. Comp. Bio.* 42: 780–789.
4. Usually, a few species are the important players in establishing the zonation patterns of rocky coasts; often, only one acts as a keystone species influencing the entire community. The textbook reviews several examples of zonation.
5. Natural disasters, and occasionally disasters caused by human blunders, can offer unique opportunities for a posteriori experiments. The *Torrey Canyon* oil spill of 1968 along the shore of southern England is an example. In killing off the molluscan grazers, this disaster showed that the upper physical limit of the intertidal algae was normally lowered by grazing.

6. Grazing is a dominant feature structuring the algal community in the rocky intertidal zone. However, algae do not lack defensive strategies. For example, under certain circumstances, algae can grow so fast as to exclude grazing gastropods. Algae also possess chemical and life-history defenses. The deposition of calcium carbonate in tissues and the production of noxious chemicals, including even sulfuric acid, can reduce grazing rates. Low encrusting growth forms in regions where grazing is high, and upright growth (frondose) where grazing is low, are examples of life-history strategies in response to grazing pressure. In a sense, these methods demonstrate *r*- and *K*-selection within a single species.
7. In general, grazing is more important in the tropics and declines in the higher latitudes, but there are additional regional patterns that are superimposed on this generalization. The cliché is still completely valid here: More research is needed!
8. Larval recruitment is a field that has received much attention. This work has been driven by both curiosity (as to how certain distribution patterns of newly settled organisms develop) and the practical aspects that these organisms foul ships and other marine structures. Larval recruitment, together with the other two biotic agents, predation and competition, can explain much about the patterns of sessile organisms on the rocky coast. Basically, larval recruitment involves a combination of habitat-dependent settlement by larvae (substratum selection behavior) followed by habitat-dependent survival (via competition, predation, and abiotic forces) of those larvae. Depending on the locale, one force or the other may dominate.
9. Large-scale patterns of zonation may be attributed to such forces as exposure and other abiotic features, as well as the biotic forces of predation, competition, and larval recruitment. Small-scale forces muddle the larger background patterns, providing texture in the local distribution of organisms. Slope, territoriality, texture (crevices), wave action, cloning, and succession are all involved in the production of small-scale patterns of distribution.
10. The age structure of invertebrates is often skewed by stochastic processes: In certain years, there is extraordinarily good (or poor) larval recruitment, or the conditions subsequent to settlement provide for increased (or decreased) survival.
11. To many people, tide pools exemplify the rocky intertidal zone. For some organisms, they represent refuges from some stressful physical factors (for example, desiccation), but for other organisms they represent elevated risk (for example, high temperature, salinity, and oxygen concentration).

Benson, K.R. 2002. The study of vertical zonation on rocky intertidal shores—a historical perspective. *Integr. Comp. Bio.* 42: 776–779.

Little, C., and J.A. Kitching. 1996. *The biology of rocky shores*. London: Oxford University Press.

Menge, B.A., et al. 1999. Top-down and bottom-up regulation of New Zealand rocky intertidal communities. *Ecol. Monog.* 69: 297–300.

Sanford, E. 2002. Water temperature, predation, and the neglected role of physiological rate effects in rocky intertidal communities. *Integr. Comp. Bio.* 42: 881–891.

12. Cobble beaches are found where recent glacial activity has deposited rocks that have not become silted in with sediments. Such habitats possess simple communities that are adapted to periodic disturbance from storms and thermal insults (both heat and cold). These habitats can become modified if stands of *Spartina alterniflora* become established. (Halophytes sometimes become established closer inshore once the *Spartina* stands are stable.) These beds then facilitate the establishment of a mussel reef community.

IV. Sandy Shores

Wave action, slope, and particle size are all correlated. Thus, knowing these physical parameters can explain much about the biota present on the beach. This section is composed of six parts:

- Environmental Conditions
- Adaptations of Organisms
- Types of Organisms

- Feeding Biology
- Community Organization on Exposed Sandy Beaches
- Community Organization on Protected Sand Flats

anoxic zone	filter feeders	refuge
backwash	fine grained	sand flats
bar	funnel feeders	scavengers
beach face	gradient	semelparous
beach profile	grain size	slope
berm	infauna	substrate movement
capillary action	infralittoral zone	supralittoral zone
coarse grained	iteroparous	surf zone
competitive gregarious behavior	midlittoral zone	swash
detritus feeders	mucous net	transition zone
diel rhythms	particle size	trophic group amensalism
dissipative beach	particle sorting	turbulent zone
dune	patches	water retention
endogenous activity	protected beach	wave action
epifauna	redox	weight belt
exposed beach	reflective beach	wrack

1. Particle size is important to burrowing organisms because of its relationship to the water held between grains. For example, finer grained sands hold more water above the tide line. Waves constantly move and sort sand grains on a beach. This phenomenon is called wave-induced substrate movement. Depending on how and where the waves break, the outcome may be beaches with predominantly fine-grained sediments or beaches with predominantly coarse-grained sediments.
2. Sandy beaches and sand flats are obviously different from rocky coasts, but some of these differences should be emphasized. Sand habitats are rather uniform in profile, but they do change seasonally with storms. While there are also crevices that act as refugia, sand is a relatively simple medium into which most organisms may burrow. Particle size is often sorted up the slope of the beach, horizontally along the beach, and vertically into the sediment. Concomitant with this physical sorting of sand is a zonation of organic matter within the vertical column. Oxygen content also is not uniform within the profile of a beach. The transition zone from oxygenated to anoxic sediments, especially in muds (see the next section), can be quite abrupt (< 1 cm), the oxygenated layer being a much lighter color than the anaerobic region. The latter region is black in color because of the precipitation of FeS.
3. The abilities to burrow within the sediment and to remain there are important features of organisms that inhabit beaches; many different adaptations correlate with these abilities. One interesting phenomenon related to burrowing is thixotropy. Thixotropy is the physical phenomenon of a gel becoming liquid when pressure is applied or when the liquid is stirred. When saturated sand (a gel) is agitated by a burrowing organism, the sand acts as a liquid, and the excavation is accomplished with much less energy than would otherwise be expected. One can observe this phenomenon by standing at the water's edge and repeatedly tapping one's foot on the sand. The result is a liquefaction of the fine sediment. This phenomenon helps in the digging and burrowing activities of many animals (for example, clams and the horseshoe crab, *Limulus*). Thixotropy may also be demonstrated with a bucket of fine, wet sand. (N.B.: Just the opposite phenomenon is seen when pressure is applied to a thick mixture of corn starch—that is, when it is squeezed.)
4. Primary production within the upper few centimeters of sediments in sandy beaches and sand flats is very minor. Here, herbivorous animals use phytoplankton and detritus as their main sources of energy. Suspension and several types of deposit feeding are common industries for these animals. Carrion feeders also may be present.
5. Zonation on sandy beaches and sand flats is present, but these patterns are not nearly as clear-cut or as well understood as those seen in the rocky intertidal zone.

V. Muddy Shores

Sandy and muddy shores occupy opposite ends of a continuum in which particle size of the sand and percent mud content vary. Sandy shores have relatively coarse-grained sand, while muddy shores have fine-grained sand. This section is divided into five parts:

- Physical Factors
- Adaptations of Organisms
- Types of Organisms
- Feeding Biology and Trophic Structure
- Zonation and Community Structure

aerobic respiration	detritus	oxygen tension
ammonia	estuaries	redox potential (Eh)
anaerobic respiration	gallery	redox-potential discontinuity (RPD)
anaerobiosis	harbors	reduction
anoxic zone	hydrogen sulfide (H ₂ S)	salt marshes
bays	lagoons	sulfur bacteria
chemoautotrophic	methane	suspension feeding
chemolithoautotrophic	mud flats	zonation
deposit feeding	oxidation	

1. Except perhaps for the aerobic–anaerobic transition, sharp boundaries do not exist in the muddy shore.
2. On the muddy shore, wave action is of minimal importance. Here, the topography is flat (mud flats), and as a result, organisms can establish more or less permanent homes; horizontal zonation, while evident in mud flats, is not particularly pronounced.
3. Just a few centimeters or less below the surface of the mud flat, the sediments go completely anaerobic (oxygen concentration = zero). Another measure of anaerobic conditions is the redox potential, or Eh, which is measured in mV. A positive Eh in the neighborhood of about 500 indicates that oxygen is available, however low its concentration may be. The region where Eh drops rapidly is called the redox-potential discontinuity (RPD). Eh values below 300 mV indicate increasing anaerobic conditions, with increasing concentrations of products of anaerobic decomposition.
4. Animals that live in mud flats construct burrows that permit entrance of oxygen-rich surface waters. Physiologically, these organisms are capable of withstanding low-oxygen tensions. One adaptation involves a modified hemoglobin molecule capable of becoming saturated with oxygen at very low levels (Fig. 6.52).
5. The RPD is a consequence of bacterial decomposition of the abundant organic matter present in this habitat. However, the layering has the additional consequence of providing two habitats for 1° production in the sediments: at the surface by algae, and at deeper levels where bacterial chemosynthesis occurs.
6. Mud flats offer a rich food supply for deposit and suspension feeders. Deposit feeders (for example, *Arenicola*, *Capitella*, and *Macoma*) are more common in habitats dominated by fine-grained sediments.

VI. Intertidal Fishes

The fish of the rocky intertidal zone are dominated by visual predators. On mud flats and sandy beaches, skates, rays, and various flatfishes prey by excavating the sediment in search of infaunal animals. This activity leaves behind depressions or pits, which lead to secondary effects on the infauna.

demersal fish	infauna	skates and rays
excavating pits	intertidal	swim bladders
flatfishes	life history	visual carnivores

VII. Birds

Many predatory birds forage along sandy and muddy shores. They partition the resources of these habitats by exhibiting selective foraging patterns. Predation on intertidal invertebrates by birds can be very intense, playing an important role in structuring these communities.

bill length	guano	raptors
bird islands	gulls	resource partitioning
dowitchers	herons	sanderlings
ducks	knots	semipalmated plovers
dunlins	oystercatcher	terns

ENRICHMENT EXERCISES

Limulus Amebocyte Lysate Assay

Outside the cell wall of gram-negative bacteria is another cell membrane called the lipopolysaccharide, or LPS. It possesses toxic properties. Because these toxins are associated with a cellular component, they are termed endotoxins. One important type of sewage contamination of water supplies is caused by the Shiga toxin, an endotoxin produced by species of the genus *Shigella*. *Escherichia coli* O157: H7, the pathogenic strain of that otherwise friendly symbiotic bacteria of our guts, is another example of an entotoxic species. The disease state is called shigellosis, bacillary dysentery, or hemorrhagic *E. coli*, depending on the bacteria in question. The aforementioned strain of *E. coli*, whose designation indicates the O antigen (the LPS toxin) and the H antigen (flagella), apparently picked up the gene for the toxin from *Shigella*.

Toxins released from the cell are called exotoxins; examples include botulin, which causes botulism, and tetanospasmin, which causes tetanus. Endotoxins from gram-negative bacteria cause fevers, diarrhea, and shock and therefore must be excluded from any agent that is to be administered intravenously. Apparently, this is no mean task. Another place where gram-negative bacteria can cause serious problems is, of course, in infections, especially of the nervous system. The disease spinal meningitis is one example of a gram-negative infection that can lead to an acute medical crisis demanding immediate intervention. In children through young adults, the suspect pathogens of spinal meningitis are often *Neisseria meningitidis* (gram negative) and *Streptococcus pneumoniae* (gram positive).

The problem for the physician is one of rapid diagnosis: Is it a gram-positive or a gram-negative bacteria that is responsible, or does the condition have a viral etiology? If gram-negative bacteria or their products (endotoxins) can be detected in the spinal fluid, then a specific course of treatment is appropriate. While standard collection and culture techniques are much too slow to isolate and grow bacteria, the *Limulus* amebocyte lysate assay (LAL test) works in a matter of an hour or so. This assay, described next, also demonstrates whether endotoxins are present in pharmaceuticals.

The blood of the horseshoe crab, *Limulus polyphemus* (Arthropoda, Chelicerata, Xiphosura), is harvested by needle puncture of the animal in the soft joint between the prosoma and opisthosoma. It is reported that the animals are not killed by this blood collection and are returned to their habitat afterwards. However, to my knowledge, their postoperative survival rate has not been determined. In the LAL test, amebocytes (or a purified aqueous lysate) from the *Limulus* blood are exposed to the suspect pharmaceutical or the patient's spinal fluid. If endotoxins from gram-negative bacteria are present in the fluid, they will cause a change in the internal structure of the amebocytes that includes degranulation, lysis, and clotting. The clotting reaction of the aqueous lysate ends in the formation of a cloudy gel. The reaction involves a stimulation of enzymes by the LPS. These enzymes then cleave a clotting protein, coagulogen, into coagulin. Changes in the test solution can be measured spectrophotometrically, with a phenomenal sensitivity [ca. picograms (10^{-12} grams) of LPS per mL solution]. Thus, when the LAL test is used, live bacteria need not be cultured to indicate a gram-negative infection of the spinal fluid or contamination of a pharmaceutical.

This phenomenon was first discovered when a researcher noticed that the blood of diseased *Limulus* clots around the site of an infection. In nature, this reaction is important because it walls off the site of gram-negative

bacterial infection. General information about this assay is available in many college-level microbiology textbooks or from a commercial supplier.

Armstrong, P.B., and M.T. Armstrong. 2003. The decorated clot: binding of agents of the innate immune system to the fibrils of the *Limulus* blood clot. *Biol. Bull.* 205: 201–203.

Isakova, V., and P.B. Armstrong. 2003. Imprisonment in a death-row cell: the fates of microbes entrapped in the *Limulus* blood clot. *Biol. Bull.* 205: 203–205.

Lim, D. 1998. *Microbiology*, 2d ed. Boston: WCB/McGraw-Hill.

Madigan, M.T., J.M. Martinko, and J. Parker. 2000. *Brock biology of microorganisms*, 9th ed. Upper Saddle River, NJ: Prentice Hall.

Nester, E.W., et al. 2001. *Microbiology: a human perspective*, 3d ed. Boston: McGrawHill.

Novitsky, T.J. (ed.). *LAL update*. Woods Hole, MA: Associates of Cape Cod, Inc. <http://www.acciusa.com/> [active as of April 16, 2004].

Walls, W.A., and J. Berkson. 2003. Effects of blood extraction on horseshoe crabs (*Limulus polyphemus*). *Fish. Bull.* 101: 457–459.

Other Aspects of *Limulus polyphemus*

For interesting reading about *Limulus*, consult the following works:

Bolton, M.L., and R.E. Loveland. 1987. Orientation of the horseshoe crab, *Limulus polyphemus*, on a sandy beach. *Bio. Bull.* 173: 289–298.

Carmichael, R.H., et al. 2004. Position of horseshoe crabs in estuarine food webs: N and C stable isotopic study of foraging ranges and diet composition. *J. Exp. Mar. Bio. Eco.* 299: 231–253.

Cohen, J.A., and H.J. Brockmann. 1983. Breeding activity and mate selection in the horseshoe crab, *Limulus polyphemus*. *Bull. Mar. Sci.* 33: 275–281. [Satellite males are considered.]

Earle, S.A. 1991. Sharks, squids, and horseshoe crabs—The significance of marine biodiversity. *BioScience* 41: 506–509.

Hata, D., and J. Berkson. 2003. Abundance of horseshoe crabs (*Limulus polyphemus*) in the Delaware Bay area. *Fish. Bull.* 101: 933–938.

Rudloe, A.E., and J. Rudloe. 1981. Horseshoe crabs: living fossils. *Nat. Geo.* 159: 562–572.

Ruta, V.J., F.A. Dodge, and R.B. Barlow. 1999. Evaluation of circadian rhythms in the *Limulus* eye. *Biol. Bull.* 197: 233–234.

Srimal, S., J. Kreiling, and J.P. Quigley. 1996. Antimicrobial activity of *Limulus* blood: role of C-reactive protein (CRP) and *Limulus* anti-lipopolysaccharide factor (LALF). *Biol. Bull.* 191: 300–301.

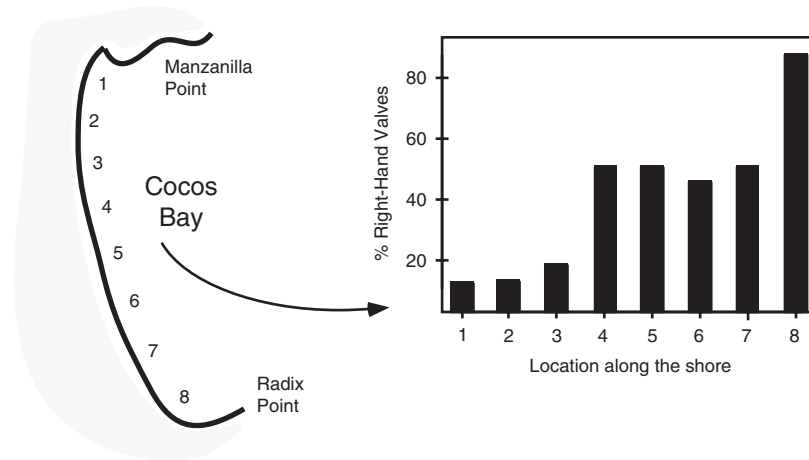
Some sample test questions about *Limulus* are as follows:

1. *Limulus* (horseshoe crab) is harvested by humans for a product isolated from its blood.
2. The LAL assay is based on a product isolated from the blood of a large intertidal arthropod.
3. The *Limulus* amebocyte lysate assay is used in medical microbiology to detect the presence of gram-negative outer-cell-membrane material (LSP) in pharmaceutical solutions and body fluids.
4. One or more satellite males often surround a mating pair of horseshoe crabs, *Limulus polyphemus*.
5. Describe the *Limulus* assay (LAL) and discuss its importance to medical microbiology. Describe how this phenomenon was first discovered.

Death Assemblages or Postmortem Shell Sorting

When walking along a beach collecting shells, one is often struck with the observation that some areas of the beach are dominated by one shell type, while in another area a different species predominates. Often the differences are not in the kind of shell, but in right- and left-hand valves! Martin-Kaye (1951) documented this

phenomenon on beaches in Trinidad. A diagrammatic representation of that work is given in the following figure:



It appears that the sorting of shells into groups by species or by left- and right-hand valves occurs because waves and longshore currents act on the shells in slightly different ways, depending on shell morphology. The result of this phenomenon is very impressive.

Kidwell, S.M. 2001. Preservation of species abundance in marine death assemblages. *Science* 294: 1091–1094.

Martin-Kaye, P. 1951. Sorting of lamellibranch valves on beaches in Trinidad, B.W.I. *Geol. Magazine* 88: 432–434.

Other Ideas Regarding Intertidal Organisms

See also similar suggestions in Chapter 8.

Chaetopterus variopedatus, the parchment tube worm, is found in mud flats along the East and Gulf Coasts. Potential activities are to observe its bioluminescence, its tube-building behavior, its feeding activities, and the interactions of the polychaete with the commensal crabs, *Pinnixa chaetopterana*.

Have students watch the reburrowing activities in clams such as *Ensis*, *Donax*, and *Mercenaria*. How do the clams go about it? Does sediment composition (that is, grain size) influence reburrowing?

Have students examine bivalve shells for evidence of predatory events—for example, drill holes on small clams or whelk (*Busycon perversum*), or pry marks on *Mercenaria mercenaria*.

Do sea cucumbers eviscerate when attacked (say, with a sharp probe), as it is widely reported in zoology textbooks? I have found that some will and some will not.

The aggressive predator *Ocypode quadrata*, the ghost crab, digs a burrow and will patrol a home range about that site. The evidence of this patrolling may be seen the next morning by examining the tracks leading to and from the burrow. Tracks leading more than two meters from the burrow may be seen. Have your students map out the previous night's patrol, looking for evidence of encounters with other animals. Is there evidence of predatory activity? Did two or more ghost crabs meet?

Some Classical, or Just Plain Interesting, References on the Intertidal Zone

Benedetti-Cecchi, L. 2000. Predicting direct and indirect interactions during succession in a mid-littoral rocky shore assemblage. *Ecol. Monogr.* 70: 45–72.

Carson, R. 1955. *The edge of the sea*. Houghton Mifflin Company.

Decho, A.W. 2000. Microbial biofilms in intertidal systems: an overview. *Cont. Shelf Res.* 20: 1257–1273.

Fleming, C.B. 1990. A snail's landscape: unlikely immigrants pick clean New England's rocky coast. *Sea Frontiers* 36: 50–55.

- Ilan, M. 1995. The life of a sponge in a sandy lagoon. *Biol. Bull.* 189: 363–369.
- Kicklighter, C.E., J. Kubanek, and M.E. Hay. 2004. Do brominated natural products defend marine worms from consumers? Some do, most don't. *Limno. Oceanogr.* 49: 430–441.
- Maddox, J. 1993. Endless ripples on the sands of time. *Nature* 364: 385.
- Menge, B.A. 1995. Indirect effects in marine rocky intertidal interaction webs: patterns and importance. *Ecol. Monogr.* 65: 21–74.
- Paine, R.T. 1994. Marine rocky shores and community ecology: an experimentalist's perspective. *Excellence in ecology*, Volume 4. *Ecol. Inst.*, Oldendorf/Luhe, Germany.
- Palmer, J.D. 1995. *The biological rhythms and clocks of intertidal animals*. New York: Oxford University Press.
- Stilgoe, J.R. 1994. *Alongshore*. New Haven, CT: Yale University Press.

Fractals and Rocky Shores

A term project for the more aggressive and mathematically minded student would be to compare the fine-scale fractal dimensions of a small section (ca. 1–3 m) of a rocky coast with and without the encrusting organisms, or to compare the fractal dimensions of two different rocky communities. On a large scale, the same comparison might be done for a kelp forest. The information would serve only to satisfy curiosity unless it could be related to some biological processes such as density of a motile predator, frequency of patch opening, rate of patch closure, change in sedimentation rate, density of species using refugia, and so on. The whole idea here is, of course, to challenge the gifted student with a different way of thinking. Some references on fractals are presented as follows:

- Abraham, E.R. 2001. The fractal branching of an arborescent sponge. *Mar. Bio.* 138: 503–510.
- Barnsley, M.F. 1993. *Fractals everywhere*, 2d ed. Cambridge, MA: Academic Press Professional.
- Ferguson, S.H., et al. 1998. Fractals, sea-ice landscape and spatial patterns of polar bears. *J. Biogeogr.* 25: 1081–1092.
- Kilps, J.R., B.E. Logan, and A.L. Alldredge. 1994. Fractal dimensions of marine snow determined from image analysis of in situ photographs. *Deep-Sea Research—Part I: Oceanographic Research Papers* 41: 1159–1169.
- Pennycuik, C.J., and N.C. Kline. 1986. Units of measurement for fractal extent, applied to the coastal distribution of bald eagle nests in the Aleutian Islands, Alaska. *Oecologia* 68: 254–258.